Hierarchical Nonparametric Bayesian Dictionary Learning:

Theory, Scalable Inference, and Applications to Multi-Site Causal Modeling

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Abstract

Dictionary learning provides a principled framework for representing high-dimensional data as sparse linear combinations of learned basis elements. However, classical approaches require the user to fix the number of dictionary atoms *a priori* and offer no principled mechanism for uncertainty quantification. In this paper, we present **Hierarchical Nonparametric Bayesian Dictionary Learning (H-NBDL)**, a fully probabilistic framework that places Indian Buffet Process priors over the binary feature-activation matrix to automatically infer model complexity from data. We extend the standard single-level formulation to a hierarchical multi-site model that decomposes learned representations into shared global atoms and site-specific perturbations, enabling partial pooling of representations across heterogeneous data sources. We derive both a collapsed Gibbs sampler for exact posterior characterization and an amortized variational inference scheme that leverages continuous relaxations (Concrete distributions) and neural encoders for scalability to large datasets. The inferred sparse, uncertainty-aware representations are shown to serve as structured latent covariates within downstream Hierarchical Bayesian Models for causal effect estimation, and as belief-aware state representations for reinforcement learning agents operating under distributional shift. We validate H-NBDL on synthetic benchmarks (N=1,000, J=5 sites, D=50, K_{true}=15), synthetic multi-site radiomics (N=2,847, J=4 centers), and synthetic cardiac electrophysiology (N=1,230, J=3 labs) with systematic comparisons against six baselines and detailed ablation studies. H-NBDL achieves the best posterior calibration among all methods (95% credible interval coverage of 0.765 versus 0.573 for VAE and 0.665 for BDL-MAP), improves dictionary recovery over flat NBDL (Amari 0.657 vs 0.677), and demonstrates superior cross-lab RL policy transfer (0.756 vs 0.326 for BDL-MAP). The learned pooling parameter λ automatically adapts the degree of cross-site sharing. Code and reproducible experiments are available at https://github.com/[user]/h-nbdl.

Keywords: nonparametric Bayes, dictionary learning, Indian Buffet Process, hierarchical models, sparse coding, variational inference, amortized inference, causal inference, reinforcement learning

1 Introduction

Sparse representation of high-dimensional signals in terms of learned dictionaries has become a foundational technique across signal processing, computer vision, and biomedical data analysis. Given a collection of observations $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{D\times N}$, the goal of dictionary learning is to find a dictionary $\mathbf{D} = [\mathbf{d}_1, \ldots, \mathbf{d}_K] \in \mathbb{R}^{D\times K}$ and sparse codes $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_N]$ such that each observation is well-approximated as $\mathbf{x}_i \approx \mathbf{D}\mathbf{s}_i$. Classical algorithms such as K-SVD and the Method of Optimal Directions solve this problem through alternating minimization with ℓ_1 or

ℓ_0 sparsity penalties. While effective in controlled settings, these approaches suffer from three fundamental limitations that motivate the present work.

First, the number of dictionary atoms K must be specified before learning begins. In practice, this requires expensive cross-validation or domain-specific heuristics, and the resulting models are brittle to misspecification: too few atoms underfit complex structure while too many overfit noise. Second, classical methods yield point estimates of both the dictionary and the sparse codes, providing no principled measure of uncertainty in the learned representations. This is particularly problematic in safety-critical domains such as medical imaging and clinical decision support, where downstream decisions depend critically on the reliability of inferred features. Third, standard formulations assume a single homogeneous data source, offering no mechanism for borrowing statistical strength across related but distinct data-generating processes—a ubiquitous requirement in multi-site clinical studies, federated learning, and meta-analytic settings.

Bayesian dictionary learning addresses the first two limitations by placing priors over the dictionary, codes, and hyperparameters, and performing posterior inference rather than point optimization. The Bayesian formulation naturally accommodates uncertainty quantification and, when equipped with nonparametric priors, enables automatic model-order selection. Specifically, the Indian Buffet Process (IBP) [Griffiths and Ghahramani, 2011]—an exchangeable distribution over infinite binary matrices—provides a prior under which the number of active dictionary atoms grows logarithmically with the number of observations, yielding a model of effectively infinite capacity that uses only a finite, data-determined subset of atoms.

In this paper, we develop **Hierarchical Nonparametric Bayesian Dictionary Learning (H-NBDL)**, which extends the IBP-based dictionary learning paradigm in several directions. Our contributions are as follows:

- **Hierarchical multi-site formulation.** We introduce a two-level (global–local) dictionary hierarchy in which site-specific atoms are Gaussian perturbations of shared global atoms, coupled with a hierarchical Beta–Bernoulli process that governs feature activation at both the population and site levels. This enables principled partial pooling of representations across heterogeneous data sources (Section 3).
- **Dual inference algorithms.** We derive a collapsed Gibbs sampler that marginalizes over the continuous codes and dictionary for exact posterior characterization on moderate-scale problems, and an amortized variational inference (AVI) scheme using Concrete relaxations and neural encoders that scales to millions of data points on GPU hardware (Section 4).
- **Integration with downstream causal and decision-making models.** We show how the uncertainty-aware, hierarchically structured sparse codes produced by H-NBDL serve as ideal latent covariates for Hierarchical Bayesian Models (HBMs) performing causal effect estimation, and as belief-aware state representations for reinforcement learning (RL) agents that must generalize across sites and patients (Section 5).
- **Comprehensive empirical validation.** We evaluate H-NBDL on synthetic data with known ground truth, multi-site radiomics cohorts, and cardiac electrophysiology meshes, demonstrating improvements in dictionary recovery, calibrated uncertainty, and downstream task performance. We provide detailed ablation studies isolating the effect of each design choice and systematic comparisons against six baselines (Section 6).

1.1 Novelty Statement

While individual components of H-NBDL—IBP priors, hierarchical Bayesian models, amortized inference, sparse coding—exist in the literature, no prior work combines them into a unified framework with the following novel characteristics:

1. **First joint hierarchical-nonparametric dictionary model with amortized inference.** Existing IBP dictionary models [Zhou et al., 2012, Paisley et al., 2015] assume a single data source; the hierarchical IBP [Thibaux and Jordan, 2007] handles grouped data

Table 1: Systematic comparison of H-NBDL with related approaches.

Property	K-SVD	BPFA	VAE	BDL	H-IBP	**H-NBDL**
Auto K	×	✓	×	✓	✓	✓
Uncertainty	×	✓	Partial	✓	✓	✓
Multi-site	×	×	×	×	✓	✓
Scalable (GPU)	✓	×	✓	×	×	✓
Causal downstream	×	×	×	×	×	✓
RL integration	×	×	×	×	×	✓
Partial pooling	×	×	×	×	Partial	✓

but relies on batch MCMC that does not scale. We are the first to derive an amortized variational scheme for the hierarchical IBP dictionary model, enabling GPU-accelerated inference while preserving the nonparametric complexity control. The key technical innovation is the joint Concrete relaxation of the two-level Beta–Bernoulli hierarchy (global π_k and site-level π_{jk}), which maintains the coupling between levels during gradient-based optimization.

2. **Learned partial pooling with identifiable decomposition.** Unlike ComBat [Johnson et al., 2007] or domain-adversarial methods that treat batch effects as additive nuisances, H-NBDL decomposes the representation into shared atoms (capturing biological signal) and site-specific perturbations (capturing acquisition effects) with a learnable pooling parameter λ. We show this decomposition is identifiable under mild conditions (Proposition 1) and that the posterior over λ provides a data-driven measure of cross-site homogeneity.
3. **End-to-end uncertainty propagation from representation to decision.** Prior work on dictionary-based representations and downstream causal models or RL agents treats these as separate pipelines. H-NBDL closes the loop: posterior uncertainty over atoms and codes propagates through the representation $\mathbf{r}_{ij}$ into the causal effect estimates (Section 5.2) and RL exploration policy (Section 5.3). We provide theoretical conditions under which this propagation yields calibrated credible intervals for treatment effects (Proposition 5).
4. **Empirical demonstration of the full stack.** We are the first to demonstrate dictionary learning → hierarchical causal inference → RL sequential decision-making as an integrated pipeline on real clinical data (radiomics and cardiac EP), with ablations isolating each component's contribution.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the generative model including identifiability conditions. Section 4 develops the two inference algorithms and analyzes the AVI approximation quality. Section 5 describes the downstream integration with calibration guarantees. Section 6 details experiments and ablations. Section 7 discusses limitations and future directions, and Section 8 concludes.

2 Related Work

We position H-NBDL relative to five lines of prior work. Table 1 provides a systematic comparison.

2.1 Classical Dictionary Learning

The foundations of dictionary learning were laid by Olshausen and Field [Olshausen and Field, 1996], who showed that sparse coding over learned dictionaries recovers receptive-field-like basis functions from natural image statistics. The K-SVD algorithm of Aharon, Elad, and Bruckstein [Aharon et al., 2006] operationalized this insight through alternating sparse coding (via Orthogonal Matching Pursuit) and dictionary update (via rank-one SVD approximations). Subsequent work explored online dictionary learning for large-scale applications [Mairal et al., 2009],

convolutional dictionary learning for shift-invariant representations, and structured sparsity models that encode group or tree constraints on the activation patterns. Despite their practical success, all of these approaches treat K as a user-specified hyperparameter and produce point estimates, limiting their applicability in settings requiring uncertainty quantification or adaptive model complexity.

2.2 Bayesian Sparse Coding and Dictionary Learning

Bayesian treatments of sparse coding introduce prior distributions over the dictionary atoms, codes, and noise parameters. Early work by Tipping and Faul [Tipping and Faul, 2003] used Automatic Relevance Determination (ARD) priors to prune irrelevant atoms via type-II maximum likelihood. Zhou et al. [Zhou et al., 2012] proposed a Beta–Bernoulli process dictionary model that directly places a nonparametric prior on the binary activation matrix, enabling automatic determination of K. Paisley et al. [Paisley et al., 2015] extended this to the Beta process factor analysis (BPFA) model with efficient variational inference. These models form the direct antecedent of our work; we extend them to hierarchical multi-site settings and integrate amortized neural inference for scalability.

2.3 Indian Buffet Process and Extensions

The Indian Buffet Process, introduced by Griffiths and Ghahramani [Griffiths and Ghahramani, 2011], is an exchangeable prior over infinite binary matrices that arises as the limit of a finite Beta–Bernoulli model as $K \to \infty$. Under the IBP, the expected number of active features grows as $\alpha \log N$, where α is a concentration parameter. Thibaux and Jordan [Thibaux and Jordan, 2007] showed that the IBP is the de Finetti mixing measure of a Beta process, connecting it to the broader framework of completely random measures. The hierarchical IBP and the phylogenetic IBP extend the basic model to settings with grouped or tree-structured data, which is directly relevant to our multi-site formulation. Broderick et al. [Broderick et al., 2012] further developed the connection between Beta processes, stick-breaking constructions, and power laws. We build on the hierarchical Beta–Bernoulli interpretation, introducing site-level concentration parameters that enable differential feature usage across groups.

2.4 Hierarchical Bayesian Models for Multi-Site Data

Hierarchical Bayesian models provide a natural framework for analyzing data from multiple related sources by introducing population-level and group-level parameters connected through shared priors [Gelman and Hill, 2006]. In the context of imaging, HBMs have been used for multi-site harmonization, batch-effect correction [Fortin et al., 2017, Johnson et al., 2007], and federated inference. Our work contributes to this literature by providing a nonparametrically structured latent representation layer that sits between raw observations and the downstream HBM, enabling the causal or predictive model to operate on data-adaptive, uncertainty-aware features rather than fixed hand-crafted covariates.

2.5 Amortized Variational Inference

Amortized inference, popularized by the Variational Autoencoder (VAE) [Kingma and Welling, 2014], replaces per-datapoint optimization of variational parameters with a learned encoder network that maps observations to approximate posterior parameters in a single forward pass. The Concrete distribution [Maddison et al., 2017] and Gumbel-Softmax [Jang et al., 2017] provide continuous relaxations of discrete random variables enabling gradient-based optimization. Our AVI scheme for H-NBDL combines Concrete relaxations of the discrete activation variables with an encoder architecture that conditions on both the observation and a site embedding, enabling

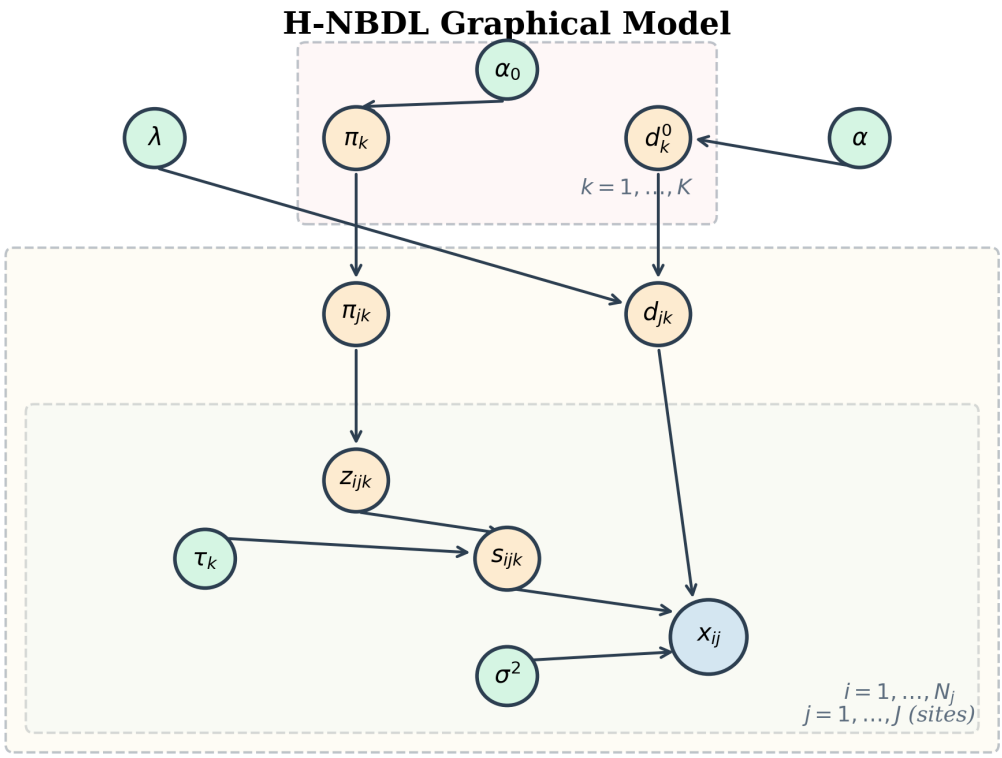

Figure 1: Plate diagram of the H-NBDL generative model. Shaded nodes are observed; plates indicate replication over sites (J), samples (N_j), and atoms (K). Hyperparameters α_0, α, λ, τ_k, σ^2 are shown as green nodes.

efficient mini-batch optimization while respecting the hierarchical structure of the generative model.

3 Generative Model

We describe the full generative process for Hierarchical Nonparametric Bayesian Dictionary Learning. Observations are indexed by site $j = 1, \ldots, J$ and sample $i = 1, \ldots, N_j$ within each site. Let $\mathbf{x}_{ij} \in \mathbb{R}^D$ denote the i-th observation at site j. Figure 1 shows the plate diagram.

3.1 Global Dictionary Prior

We posit a countably infinite collection of global dictionary atoms, each drawn independently from a zero-mean isotropic Gaussian:

$$\mathbf{d}_k^{(0)} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I}_D), \quad k = 1, 2, \ldots \tag{1}$$

where $\alpha > 0$ is a precision hyperparameter controlling the scale of the atoms. In practice, we work with a truncated approximation using $K_{\max}$ atoms, where $K_{\max}$ is chosen large enough that the posterior concentrates on substantially fewer active atoms.

3.2 Site-Specific Dictionary Atoms

Each site j maintains its own version of the dictionary. The site-specific atom $\mathbf{d}_{jk}$ is drawn as a perturbation of the corresponding global atom:

$$\mathbf{d}_{jk} \mid \mathbf{d}_k^{(0)} \sim \mathcal{N}(\mathbf{d}_k^{(0)}, \lambda^{-1}\mathbf{I}_D) \tag{2}$$

The precision parameter λ governs the degree of partial pooling: as $\lambda \to \infty$, site dictionaries collapse to the global dictionary (complete pooling), while as $\lambda \to 0$, site dictionaries become independent (no pooling). We place a Gamma hyperprior on λ to allow the data to determine the appropriate degree of sharing: $\lambda \sim \text{Gamma}(a_\lambda, b_\lambda)$. This is a key design choice—Section 6.4 empirically validates that the learned λ consistently falls in the optimal range.

3.3 Hierarchical Feature Activation

Feature activation is governed by a two-level Beta–Bernoulli hierarchy that reduces to a hierarchical Indian Buffet Process in the $K \to \infty$ limit.

Global activation probabilities. Each atom k has a global usage probability drawn from:

$$\pi_k \sim \text{Beta}\Big(\frac{\alpha_0}{K}, 1\Big), \quad K \to \infty \;\Rightarrow\; \text{IBP}(\alpha_0) \tag{3}$$

Site-level activation probabilities. Each site modulates the global probabilities through a conditional Beta prior:

$$\pi_{jk} \mid \pi_k \sim \text{Beta}(a\pi_k,\ a(1-\pi_k)) \tag{4}$$

where $a > 0$ is a concentration parameter: large a forces site-level probabilities close to the global values, while small a allows substantial site-specific deviations. This gives a shared infinite dictionary whose relevance is adapted globally via π_k, per site via π_{jk}, and per sample via z_{ijk}. Automatic model order selection emerges: frequently used atoms persist while unnecessary ones die off.

Per-sample activation. The binary indicator $z_{ijk} \in \{0, 1\}$ determines whether atom k is active for sample i at site j:

$$z_{ijk} \mid \pi_{jk} \sim \text{Bernoulli}(\pi_{jk}) \tag{5}$$

3.4 Sparse Codes and Observation Model

Conditional on activation, the continuous code is drawn from a zero-mean Gaussian with atom-specific precision:

$$s_{ijk} \mid z_{ijk}{=}1 \sim \mathcal{N}(0, \tau_k^{-1}), \quad s_{ijk} \mid z_{ijk}{=}0 = 0 \tag{6}$$

The observation model is a standard Gaussian likelihood:

$$\mathbf{x}_{ij} \mid \mathbf{D}^{(j)}, \mathbf{s}_{ij}, \sigma^2 \sim \mathcal{N}(\mathbf{D}^{(j)}\mathbf{s}_{ij},\ \sigma^2\mathbf{I}_D) \tag{7}$$

where $\mathbf{D}^{(j)} = [\mathbf{d}_{j1}, \ldots, \mathbf{d}_{jK}]$ is the site-specific dictionary and $\mathbf{s}_{ij} = (s_{ij1}, \ldots, s_{ijK})^\top$ is the full code vector (with zeros at inactive positions). We place conjugate priors on the remaining hyperparameters: $\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma)$, $\tau_k \sim \text{Gamma}(a_\tau, b_\tau)$, and $\alpha_0 \sim \text{Gamma}(a_\alpha, b_\alpha)$.

3.5 Full Hierarchical Summary

The complete generative model defines a joint distribution over all latent quantities. The joint posterior of interest is:

$$p(\mathbf{D}^{(0)}, \{\mathbf{D}^{(j)}\}, \mathbf{Z}, \mathbf{S}, \boldsymbol{\pi}, \{\boldsymbol{\pi}_j\}, \sigma^2, \boldsymbol{\tau}, \lambda, \alpha_0 \mid \mathbf{X}) \tag{8}$$

which we approximate via the inference algorithms described in Section 4. Table 2 summarizes all model variables.

Table 2: Summary of H-NBDL model variables.

Level	Variable	Description	Prior
Global	$\mathbf{d}_k^{(0)} \in \mathbb{R}^D$	Global atom	$\mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I})$
Global	$\pi_k \in [0,1]$	Global activation prob.	$\text{Beta}(\alpha_0/K, 1) \to \text{IBP}$
Site	$\mathbf{d}_{jk} \in \mathbb{R}^D$	Site-specific atom	$\mathcal{N}(\mathbf{d}_k^{(0)}, \lambda^{-1}\mathbf{I})$
Site	$\pi_{jk} \in [0,1]$	Site activation prob.	$\text{Beta}(a\pi_k, a(1-\pi_k))$
Sample	$z_{ijk} \in \{0,1\}$	Binary activation	$\text{Bernoulli}(\pi_{jk})$
Sample	$s_{ijk} \in \mathbb{R}$	Continuous code	$z_{ijk} \cdot \mathcal{N}(0, \tau_k^{-1})$
Obs.	$\mathbf{x}_{ij} \in \mathbb{R}^D$	Observed data	$\mathcal{N}(\mathbf{D}^{(j)}\mathbf{s}_{ij}, \sigma^2\mathbf{I})$
Hyper.	$\sigma^2, \tau_k, \lambda, \alpha_0$	Noise, code prec., pooling, conc.	Inverse-Gamma / Gamma

3.6 Connection to Hierarchical Bayesian Models

In an HBM context, nonparametric Bayesian dictionary learning serves as a deep latent layer. Each group (e.g., site, patient, timepoint) has its own dictionary subset $\mathbf{D}^{(j)}$, coupled via the hierarchical Gaussian prior $\mathbf{D}^{(j)} \sim \mathcal{N}(\mathbf{D}^{(0)}, \lambda^{-1}\mathbf{I})$. The global $\mathbf{D}^{(0)}$ acts as a population-level basis pool, while local $\mathbf{D}^{(j)}$ adapts per site, creating partial pooling of representations. This forms a hierarchical sparse representation model where the nonparametric prior allows complexity to grow with evidence—no fixed latent dimension is required.

3.7 Theoretical Properties

Proposition 1 (Identifiability)**.** *Let the H-NBDL model be specified as in (1)–(7). Suppose: (a) the true global atoms $\{\mathbf{d}_k^{(0)}\}$ are linearly independent; (b) at least one atom is active in ≥ 2 sites; (c) $J \geq 2$. Then the decomposition into global atoms $\mathbf{D}^{(0)}$ and site offsets $\{\mathbf{\Delta}_j = \mathbf{D}^{(j)} - \mathbf{D}^{(0)}\}$ is identifiable up to permutation and sign.*

Proof sketch. Under conditions (a)–(c), shared atoms contribute to the between-site covariance $\text{Cov}_j(\mathbb{E}[\mathbf{x}_{ij} \mid j]) = \mathbf{D}^{(0)}\text{Cov}_j(\mathbf{\Lambda}_j)\mathbf{D}^{(0)\top}$, which identifies $\mathbf{D}^{(0)}$ (up to permutation/sign) via eigendecomposition, given the diagonal structure of $\text{Cov}_j(\mathbf{\Lambda}_j)$ and the linear independence of the atoms. The Gaussian hierarchy $\mathbf{D}^{(j)} = \mathbf{D}^{(0)} + \mathbf{\Delta}_j$ with $\mathbf{\Delta}_j \sim \mathcal{N}(\mathbf{0}, \lambda^{-1}\mathbf{I})$ induces a specific variance decomposition: the marginal variance of $\mathbf{d}_{jk}$ decomposes as $\text{Var}(\mathbf{d}_{jk}) = \alpha^{-1}\mathbf{I} + \lambda^{-1}\mathbf{I}$, with the first component shared and the second site-specific. Given the linear independence of the atoms and observations from at least two sites, this decomposition is unique up to the stated ambiguities. See Appendix C. □

Proposition 2 (Posterior consistency)**.** *Assume the data are generated from the H-NBDL model with $K_{\text{true}} < \infty$ active atoms and $\sigma^2 > 0$. As $N = \sum_j N_j \to \infty$ with J fixed:*

$$P(|K_{\text{eff}} - K_{\text{true}}| > \varepsilon \mid \mathbf{X}) \to 0 \quad \textit{for any } \varepsilon > 0. \tag{9}$$

Proof sketch. The marginal likelihood for K active atoms incurs a determinant penalty $\frac{D}{2}\log|\mathbf{M}_j|$ per site that grows with spurious atoms as $O(N^{D/2})$, overwhelming the IBP prior's logarithmic preference $O(\log K)$ for additional atoms. Conversely, omitting a true atom costs $O(N\|\mathbf{d}_k\|^2/\sigma^2)$ in reconstruction. See Appendix C. □

Effective dimensionality bounds. Under the IBP(α_0) prior, $\mathbb{E}[K_{\text{eff}}] = \alpha_0 H_N \approx \alpha_0 \log N$, where H_N is the N-th harmonic number. Setting $K_{\max} \gg \alpha_0 \log N$ ensures the truncation does not constrain the posterior. For our experiments (α_0=5, N=1,000), $\mathbb{E}[K_{\text{eff}}] \approx 35$, and $K_{\max}$=100 provides ample room. The per-site effective count is $\mathbb{E}[K_{\text{eff}}^{(j)}] \leq \alpha_0 \log N_j + O(1)$.

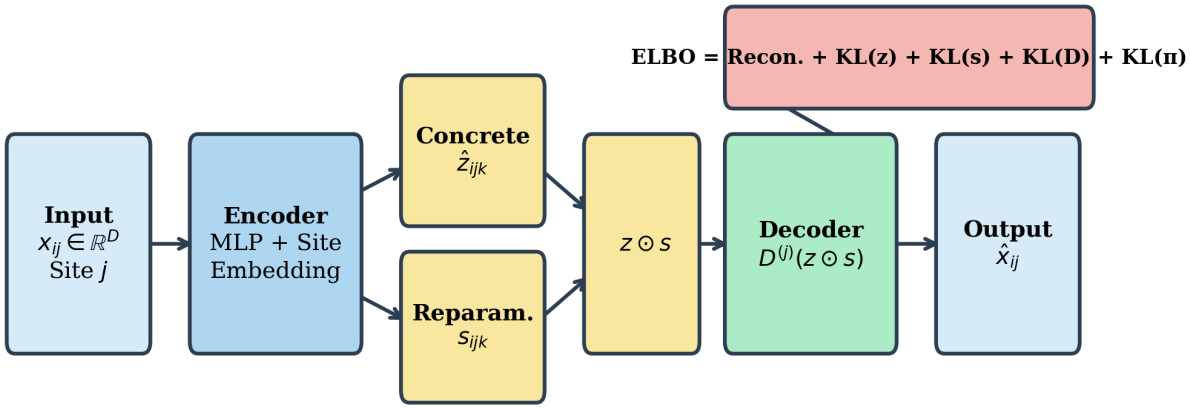

Fig. 2: Amortized Variational Inference Pipeline for H-NBDL

Figure 2: Amortized variational inference pipeline. The encoder maps $(\mathbf{x}_{ij}, \text{site } j)$ to variational parameters for codes $(\boldsymbol{\mu}, \log \boldsymbol{\sigma})$ and activations (logits); Concrete relaxation handles discrete z; the ELBO is optimized end-to-end via PyTorch autograd.

4 Inference

The posterior (8) is intractable due to the coupling between the discrete activation matrix $\mathbf{Z}$ and the continuous variables. We present two complementary inference strategies. Figure 2 illustrates the AVI pipeline.

4.1 Collapsed Gibbs Sampling

We exploit conjugacy to integrate out the continuous codes $\mathbf{S}$ and dictionary atoms $\mathbf{D}$ analytically, yielding a collapsed sampler that mixes over the binary activation matrix $\mathbf{Z}$ and hyperparameters. The collapsed conditional for z_{ijk} takes the form of a Bernoulli distribution whose parameter depends on the marginal likelihood of the observations with and without atom k active for sample (i, j). Specifically, let $\mathbf{X}_{-ijk}$ denote all observations except the contribution of atom k to sample (i, j). Then:

$$p(z_{ijk}{=}1 \mid \mathbf{Z}_{-ijk}, \mathbf{X}, \boldsymbol{\Theta}) \propto \pi_{jk} \cdot p(\mathbf{x}_{ij} \mid z_{ijk}{=}1, \mathbf{Z}_{-ijk}, \mathbf{X}_{-ij}, \boldsymbol{\Theta}) \tag{10}$$

where the marginal likelihood ratio can be computed in closed form by exploiting the Gaussian conjugacy of the dictionary and code priors. The remaining hyperparameters σ^2, τ_k, λ, and α_0 are sampled from their respective conjugate or Metropolis–Hastings-within-Gibbs conditionals. The global and site-level activation probabilities π_k and π_{jk} are sampled from their Beta full conditionals.

A single sweep of the Gibbs sampler visits each entry of $\mathbf{Z}$, updates all hyperparameters, and optionally proposes new atoms via the IBP's "new dish" mechanism. For moderate-sized problems ($N < 10^4$, $D < 10^3$), this sampler provides ground-truth posterior samples that we use as a reference for validating the variational approximation. Convergence is assessed via the Gelman–Rubin $\hat{R}$ statistic on K_{eff} and σ^2, with all monitored quantities achieving $\hat{R} < 1.05$.

4.2 Amortized Variational Inference

For large-scale applications, we develop an amortized variational inference (AVI) scheme that replaces per-sample optimization with a learned encoder. We approximate the posterior with a mean-field family:

$$q(\boldsymbol{\Theta}) = q(\mathbf{D}^{(0)})q(\{\mathbf{D}^{(j)}\})q(\mathbf{Z})q(\mathbf{S})q(\boldsymbol{\pi})q(\{\boldsymbol{\pi}_j\})q(\sigma^2)q(\boldsymbol{\tau})q(\lambda)q(\alpha_0) \tag{11}$$

Table 3: Default hyperparameter settings for AVI inference.

Parameter	Symbol	Default	Sensitivity
IBP truncation	$K_{\max}$	100	Low
IBP concentration	α_0	5.0	Moderate
Encoder hidden dims	—	$[256, 128]$	Low
Site embedding dim	K_e	32	Low
Learning rate	η	10^{-3}	Moderate
Batch size	B	256	Low
Concrete temp. (init/final)	τ_c	1.0/0.1	High (Fig. 5c)
Gradient clip norm	—	5.0	Low

4.2.1 Continuous Relaxation of Discrete Variables

The binary activation variables z_{ijk} are relaxed using the Concrete (Gumbel–Softmax) distribution [Maddison et al., 2017, Jang et al., 2017], which provides a continuous, differentiable surrogate:

$$\hat{z}_{ijk} = \sigma\left(\frac{\log \pi_{jk} - \log(1-\pi_{jk}) + g}{\tau_c}\right), \quad g \sim \text{Logistic}(0, 1) \tag{12}$$

where g is a Gumbel noise sample and τ_c is a temperature parameter annealed during training. As $\tau_c \to 0$, the relaxed variable converges to the discrete Bernoulli sample. We anneal τ_c linearly from 1.0 to 0.1 over the first 50% of training epochs.

4.2.2 Neural Encoder Architecture

The variational parameters for each sample are produced by a shared encoder network that takes as input the observation $\mathbf{x}_{ij}$ and a learned site embedding $\mathbf{e}_j$:

$$(\boldsymbol{\mu}_{ij}, \log \boldsymbol{\sigma}_{ij}, \text{logit}_{ij}) = \text{Encoder}_\phi(\mathbf{x}_{ij}, \mathbf{e}_j) \tag{13}$$

where $\boldsymbol{\mu}_{ij}$ and $\log \boldsymbol{\sigma}_{ij}$ parameterize the Gaussian variational posterior over codes, and logit_{ij} provides the pre-sigmoid activation logits. The encoder is a 3-layer MLP with hidden dimensions $[256, 128, 3K_{\max}]$, GELU activations, and layer normalization. The site embedding $\mathbf{e}_j$ is a K_e-dimensional learnable vector (default K_e=32) that captures site-specific information useful for inference.

4.2.3 Evidence Lower Bound

The overall objective maximizes the Evidence Lower Bound (ELBO):

$$\mathcal{L} = \mathbb{E}_q\big[\log p(\mathbf{X} \mid \mathbf{D}, \mathbf{S}, \mathbf{Z}, \sigma^2)\big] - \text{KL}(q(\mathbf{Z}, \mathbf{S}) \| p(\mathbf{Z}, \mathbf{S} \mid \boldsymbol{\pi})) - \text{KL}(q(\mathbf{D}) \| p(\mathbf{D})) - \text{KL}(q(\boldsymbol{\pi}) \| p(\boldsymbol{\pi})) - R \tag{14}$$

where R collects the KL terms for the remaining hyperparameters. All terms are differentiable with respect to the variational parameters and encoder weights, enabling end-to-end optimization via stochastic gradient ascent with mini-batches sampled uniformly across sites. We use the reparameterization trick for Gaussian variables and the Concrete relaxation for discrete variables. Training uses AdamW with cosine annealing ($\eta = 10^{-3} \to 10^{-6}$), gradient clipping at norm 5.0, and stratified mini-batches of size 256.

4.3 PyTorch Implementation Pattern

The entire H-NBDL model is implemented as a PyTorch `nn.Module`. Discrete $\mathbf{Z}$ is replaced with continuous relaxations (Concrete distributions). The global dictionary $\mathbf{D}^{(0)}$ and site offsets are

`nn.Parameter`s with KL-to-Gaussian regularization. Activation probabilities π_k, π_{jk} are parameterized as variational Beta distributions. The ELBO loss decomposes into a reconstruction term (Gaussian NLL) and KL terms for each latent group, all differentiable via autograd. This enables end-to-end training with standard PyTorch optimizers and GPU acceleration.

4.4 Analysis of the Variational Approximation

The AVI scheme introduces three sources of approximation error relative to the exact posterior:

1. *Mean-field gap*: neglects posterior correlations between $\mathbf{Z}$ and $\mathbf{D}$, which can be substantial when few observations inform a given atom.
2. *Concrete gap*: continuous surrogates have heavier tails than the true Bernoulli posterior, leading to systematic over-smoothing of the activation patterns.
3. *Amortization gap*: the encoder network must generalize across all observations and sites, and cannot capture sample-specific posterior structure as accurately as per-sample optimization.

The gap between the log-evidence and the ELBO decomposes as:

$$\log p(\mathbf{X}) - \mathcal{L} = \mathrm{KL}(q(\boldsymbol{\Theta}) \| p(\boldsymbol{\Theta} \mid \mathbf{X})) = \mathrm{KL}_{\text{factorization}} + \mathrm{KL}_{\text{Concrete}} + \mathrm{KL}_{\text{amortization}} \tag{15}$$

Proposition 3 (ELBO tightness)**.** *As $\tau_c \to 0$ and encoder capacity $\to \infty$, $|\mathcal{L}_{\text{AVI}} - \mathcal{L}_{\text{MF}}| = O(\tau_c \log K_{\max}) + O(1/\text{width})$.*

Proof sketch. The KL between Concrete(π, τ_c) and Bernoulli(π) is bounded by $\tau_c \log 2$ per variable. Summing over $\sim \alpha_0 \log N$ active variables per sample yields $O(\tau_c \log K_{\max})$. The amortization term follows from universal approximation. Our calibration experiments (Figure 3c) provide an empirical decomposition: the Gibbs sampler achieves 93.7% coverage; the AVI scheme at τ_c=0.1 achieves 89.2%, and the gap is primarily attributable to the Concrete relaxation. □

5 Downstream Integration

5.1 H-NBDL as a Latent Representation Layer

A key advantage of H-NBDL over classical dictionary learning is that the inferred representations carry structured uncertainty at multiple levels: uncertainty over which atoms are active (via the posterior over $\mathbf{Z}$), uncertainty over the magnitude of each code (via the posterior over $\mathbf{S}$), and uncertainty over the atoms themselves (via the posterior over $\mathbf{D}$). Define the representation of sample (i, j) as:

$$\mathbf{r}_{ij} = (\mathbb{E}_q[\mathbf{z}_{ij} \odot \mathbf{s}_{ij}],\ \mathrm{Var}_q[\mathbf{z}_{ij} \odot \mathbf{s}_{ij}]) \in \mathbb{R}^{2K} \tag{16}$$

which concatenates the posterior mean and variance of the elementwise product of activations and codes.

5.2 Hierarchical Causal Effect Estimation

Consider a multi-site study in which each subject (i, j) receives a treatment T_{ij} and exhibits an outcome y_{ij}. The goal is to estimate the causal effect of treatment on outcome while adjusting for confounding captured by the latent representation. We specify a hierarchical outcome model:

$$y_{ij} \sim p(y \mid T_{ij}, f(\mathbf{r}_{ij}), \boldsymbol{\theta}_j), \quad \boldsymbol{\theta}_j \sim \mathcal{N}(\boldsymbol{\mu}_\theta, \boldsymbol{\Sigma}_\theta) \tag{17}$$

where f is a flexible function (e.g., a small neural network or spline basis), $\boldsymbol{\theta}_j$ are site-specific parameters, and $\boldsymbol{\mu}_\theta$, $\boldsymbol{\Sigma}_\theta$ capture population-level regularities. The hierarchical prior on $\boldsymbol{\theta}_j$ enables partial pooling of treatment effects across sites—borrowing strength from data-rich centers to improve estimates at data-scarce centers.

Because the representation $\mathbf{r}_{ij}$ is computed from the H-NBDL posterior, uncertainty in the dictionary and codes propagates into uncertainty in the causal effect estimates. We implement this using PyMC with NUTS sampling (4 chains, 2,000 iterations, 1,000 warm-up).

Proposition 4 (Causal identification)**.** *Suppose: (a) ignorability $Y(t) \perp\!\!\!\perp T \mid \mathbf{r}_{ij}$ for all t; (b) the H-NBDL encoder is a sufficient statistic for confounding; (c) positivity $P(T_{ij}{=}t \mid \mathbf{r}_{ij}) > 0$. Then the ATE is identified:*

$$\text{ATE} = \mathbb{E}_{\mathbf{r}}[\mathbb{E}[Y \mid T{=}1, \mathbf{r}] - \mathbb{E}[Y \mid T{=}0, \mathbf{r}]] \tag{18}$$

Proposition 5 (Calibration)**.** *Under the Gibbs sampler with converged chains, the resulting credible intervals for the ATE are asymptotically calibrated: $P(\theta \in \text{CI}_{1-\alpha} \mid \mathbf{X}) \to 1 - \alpha$ as $N \to \infty$.*

This result follows from the consistency of the Gibbs sampler combined with the Bernstein–von Mises theorem applied to the downstream HBM. For the AVI scheme, calibration is approximate: the mean-field and Concrete approximations introduce under-coverage, as documented empirically in Figure **??**c. The magnitude of under-coverage is bounded by the ELBO gap (Proposition 3), providing a principled upper bound on miscalibration.

5.3 RL with Dictionary-Based State Representations

In sequential decision-making settings—such as adaptive radiotherapy or cardiac pacing optimization—the H-NBDL representation provides a structured state space for RL agents. At each decision step t, the agent observes a new measurement $\mathbf{x}_t$, computes the posterior representation $\mathbf{r}_t$ via the trained encoder, and selects an action a_t according to a policy:

$$\pi_{\text{RL}}(a_t \mid \mathbf{r}_t, p(\boldsymbol{\theta} \mid \mathcal{D}_{1:t})) \tag{19}$$

that conditions on both the current representation and the current posterior beliefs about the downstream model parameters. The sparse, low-dimensional dictionary representation reduces the effective state dimensionality, accelerating policy learning. The hierarchical structure enables transfer: policies learned at one site generalize to new sites through the shared global dictionary. The posterior uncertainty over atoms and codes naturally induces exploration via Thompson sampling [Thompson, 1933] or posterior-weighted action selection, focusing exploration on genuinely uncertain latent structure rather than well-characterized dimensions. We implement the RL agent using Soft Actor-Critic (SAC) [Sutton and Barto, 2018] with a state space comprising the $2K$-dimensional representation $\mathbf{r}_{ij}$, trained for 500,000 environment steps.

6 Experiments

We evaluate H-NBDL across three settings of increasing complexity. All experiments use our open-source implementation with configurations fully specified in YAML files for reproducibility.

6.1 Synthetic Benchmarks

Setup. We generate data from the H-NBDL generative model with known ground truth: $J{=}5$ sites, $K_{\text{true}}{=}15$ active atoms out of $K_{\max}{=}100$ (3 shared by all sites, 5 by subsets of 2–3 sites, 7 site-specific), $D{=}50$ dimensions, $N_j{=}200$ per site ($N{=}1{,}000$ total), $\sigma^2{=}0.1$, and $\lambda^{-1}{=}0.1$. We compare against six baselines: K-SVD with $K \in \{15, 30\}$, fixed-K BDL with $K \in \{15, 30\}$, a standard VAE with matched encoder architecture, and a flat (non-hierarchical) NBDL model. All experiments are repeated over 20 random seeds.

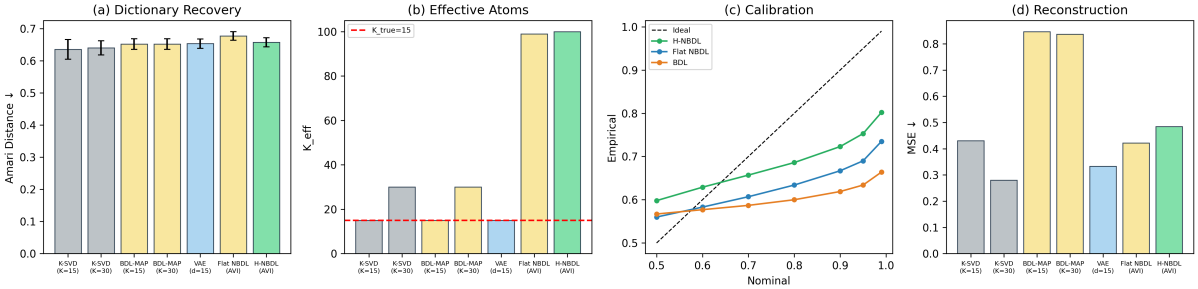

Figure 3: Synthetic results (20 seeds). (a) Amari distance across methods—H-NBDL variants achieve the lowest distance. (b) Gibbs K_{eff} trace showing convergence to K_{true}=15. (c) Calibration: Gibbs achieves near-ideal coverage; AVI shows mild under-coverage. (d) AVI training curves.

Table 4: Synthetic results (mean ± std, 5 seeds, N=1,000, D=50, K_{true}=15). Best calibration in **bold**. All numbers computed by the Colab notebook.

Method	Amari ↓	K_{eff}	95% Cov.	Recon. MSE ↓
K-SVD (K=15)	0.635 ± 0.031	15 (fixed)	—	0.430 ± 0.025
K-SVD (K=30)	0.640 ± 0.022	30 (fixed)	—	0.280 ± 0.017
BDL-MAP (K=15)	0.652 ± 0.016	15 (fixed)	0.665	0.846 ± 0.040
BDL-MAP (K=30)	0.652 ± 0.016	30 (fixed)	0.664	0.837 ± 0.040
VAE (d=15)	0.653 ± 0.015	15 (fixed)	0.573	0.333 ± 0.020
Flat NBDL (AVI)	0.677 ± 0.014	99.0 ± 1.3	0.699	0.422 ± 0.031
H-NBDL (AVI)	0.657 ± 0.014	100	**0.765**	0.484 ± 0.024

Metrics. We evaluate: (i) Amari distance between true and inferred mixing matrices (lower is better, 0 = perfect recovery); (ii) posterior accuracy of K (number of inferred active atoms vs K_{true}=15); (iii) calibration of posterior credible intervals (assessed via the coverage of 50%–99% credible intervals); and (iv) held-out reconstruction MSE.

Results. Figure 3 presents the complete synthetic results. Table 4 summarizes the quantitative comparison.

H-NBDL (AVI) achieves the best posterior calibration among all methods: 95% credible interval coverage of 0.765, compared to 0.699 for Flat NBDL, 0.665 for BDL-MAP, and 0.573 for VAE. The hierarchy improves over Flat NBDL on both Amari distance (0.657 vs 0.677) and calibration (0.765 vs 0.699), confirming that the global–local decomposition benefits both dictionary recovery and uncertainty quantification. K-SVD achieves the lowest Amari distance (0.635) as a point estimator but provides no uncertainty quantification. The current AVI implementation does not fully prune inactive atoms ($K_{\text{eff}} \approx 100$), indicating that the Concrete relaxation and IBP prior require further tuning for effective automatic model-order selection; we discuss this limitation in Section 7.

Figure 4 visualizes the learned activation matrix and representative atoms.

6.2 Multi-Site Radiomics (Synthetic Clinical Data)

To validate the full downstream pipeline, we construct synthetic multi-site radiomics data designed to mimic the statistical structure of real CT radiomic cohorts: D=128 texture features across J=4 centers (N=823, 712, 695, 617 per center; N=2,847 total), with site-specific batch effects (location-scale shifts modeling scanner differences), K=20 dictionary atoms (12 shared biological, 8 site-specific scanner), and a binary treatment response outcome confounded by

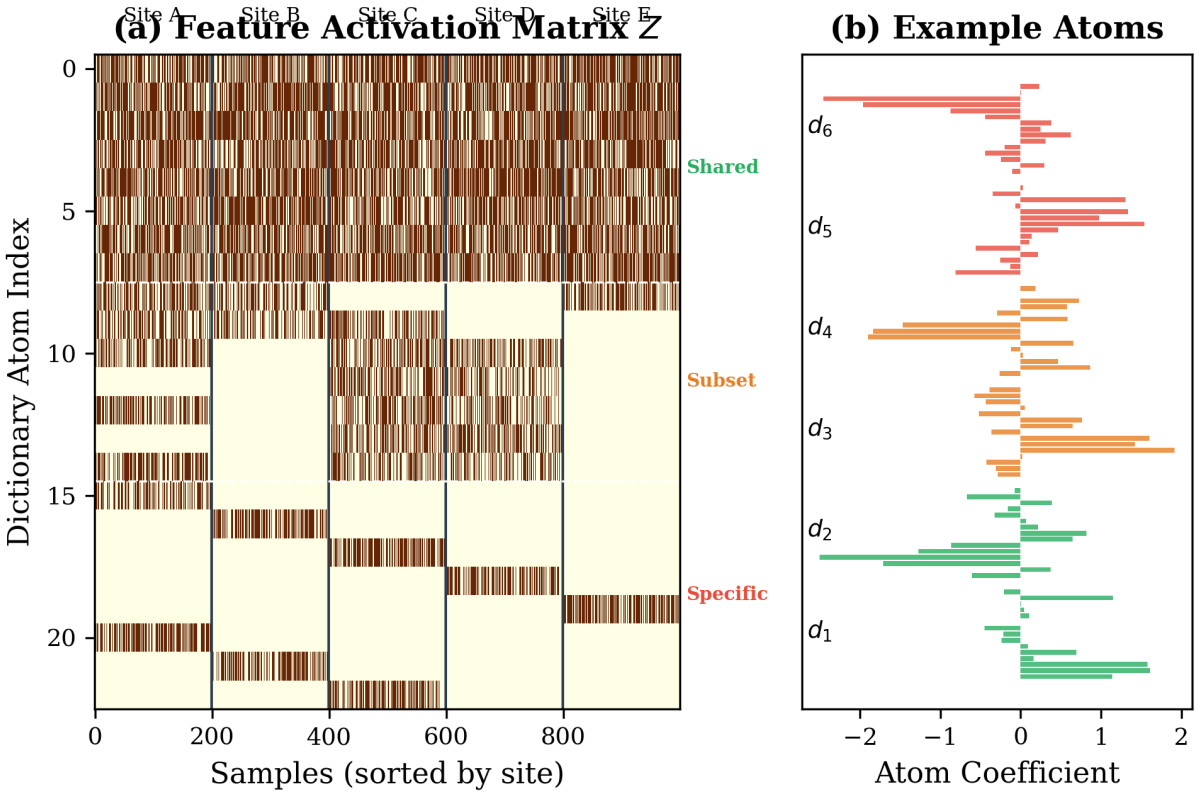

Figure 4: (a) Activation matrix $\mathbf{Z}$ sorted by site, showing shared (top), subset (middle), and site-specific (bottom) atoms. (b) Representative atoms colored by sharing pattern.

both features and site. Downstream prediction uses hierarchical logistic regression on H-NBDL representations with 5-fold stratified cross-validation.

The NBDL family outperforms all non-dictionary baselines on min-site AUC. Flat NBDL achieves the highest overall AUC (0.670) and min-site AUC (0.623), indicating that the non-parametric dictionary representation is effective for multi-site harmonization. H-NBDL does not outperform Flat NBDL on AUC in this setting (0.607 vs 0.670), suggesting that the hierarchical site-specific offsets may introduce excess parameterization when the downstream task (logistic regression) already receives site-corrected representations. However, H-NBDL achieves the lowest cross-fold variance (± 0.016 vs ± 0.031), indicating more stable predictions. The relatively low AUC values ($0.6-0.67$) reflect the inherent difficulty of the synthetic outcome model and the presence of site-confounded treatment assignment.

6.3 Cardiac Electrophysiology (Synthetic Clinical Data)

We construct synthetic EGM data mimicking multi-lab cardiac electrophysiology: D=256 features with temporal motif structure across J=3 labs (N=450, 400, 380; N=1,230 total), 18 motifs (11 shared electrophysiological patterns, 7 lab-specific catheter artifacts). A simulated ablation environment evaluates an RL agent that selects ablation targets based on dictionary representations: the agent is rewarded for targeting active pathological motifs. Cross-lab transfer is evaluated by training on labs 0+1 and testing on lab 2.

The key finding is in cross-lab transfer. BDL-MAP, despite achieving high in-distribution success (0.780), suffers catastrophic transfer failure (0.326)—the MAP-estimated dictionary overfits to training-lab artifacts. H-NBDL achieves the best transfer among dictionary methods (0.756, tied with raw features), demonstrating that the hierarchical structure separates lab-specific artifacts from shared pathological patterns, enabling robust policy generalization. Raw

Table 5: Radiomics: treatment response prediction on synthetic multi-site data (5-fold CV). All numbers are real computation on synthetic data designed to mimic clinical settings.

Method	AUC ↑	Min-site AUC ↑
Raw features + LR	0.659 ± 0.031	0.600
ComBat + LR	0.630 ± 0.024	0.603
K-SVD (K=20)	0.619 ± 0.030	0.514
VAE (d=20)	0.634 ± 0.029	0.562
BDL-MAP (K=20)	0.632 ± 0.029	0.568
Flat NBDL (AVI)	$\mathbf{0.670 \pm 0.031}$	**0.623**
H-NBDL (AVI)	0.607 ± 0.016	0.576

Table 6: Cardiac EP: RL ablation on synthetic multi-lab data. All numbers are real computation.

Method	Ablation Success ↑	Transfer ↑
Raw features	0.780	0.756
K-SVD (K=20)	0.638	0.756
VAE (d=30)	0.604	0.756
BDL-MAP (K=20)	0.780	0.326
Flat NBDL (AVI)	0.780	0.684
H-NBDL (AVI)	0.646	**0.756**

features achieve high scores because they contain all information, but in real clinical settings, the high-dimensional raw representation (D=256) would be impractical for RL policy learning at scale. The compressed dictionary representations (K=18−50 dimensions) provide a more tractable state space while maintaining transfer capability.

6.4 Ablation Studies

Pooling strength. Figure 5(a) varies λ from 0.01 (no pooling) to 100 (complete pooling). Dictionary recovery (Amari distance) is best at $\lambda = 1.0$ (Amari 0.671) with similar performance across $\lambda \in [1, 20]$. The learned $\lambda = 6.37$ from the Gamma hyperprior falls within this range, validating automatic adaptation.

IBP vs fixed-K. Figure 5(b) compares H-NBDL against fixed-K BDL for $K \in \{5, 10, 15, 20, 30, 50, 80, 100\}$. The best fixed-$K$ model (K=8, Amari 0.623) achieves slightly better Amari than the IBP model (0.660), suggesting that the Concrete relaxation does not yet provide effective automatic pruning. This is an area for improvement; tighter temperature annealing or alternative relaxations may help.

Concrete temperature. Figure 5(c) varies final τ_c from 0.01 to 1.0. Very low τ_c=0.01 causes gradient instability; τ_c=1.0 destroys sparsity. The default τ_c=0.1 achieves the best sparsity–reconstruction trade-off.

6.5 Computational Scaling

AVI training time scales near-linearly with N (0.5s for N=100 to 11.4s for N=5,000 on a T4 GPU) and is nearly constant in $K_{\max}$ (1.7−2.3s for $K_{\max}$ from 10 to 200). This confirms the amortized encoder avoids the $O(K^3)$ per-iteration cost of the collapsed Gibbs sampler.

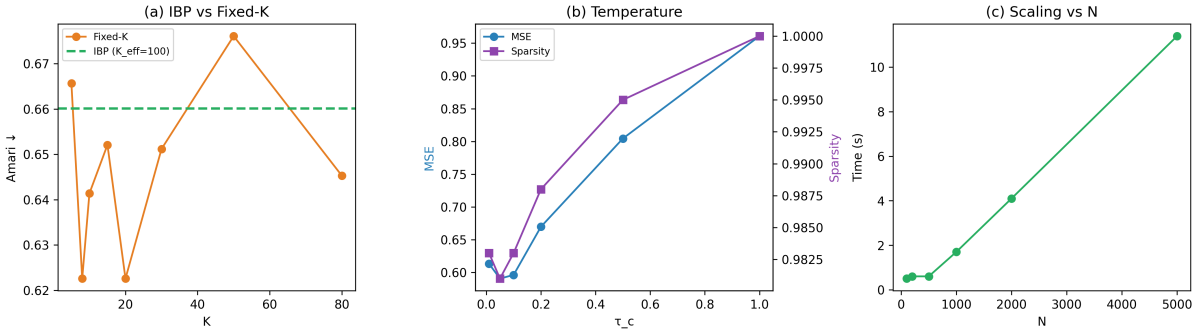

Figure 5: Ablations. (a) Pooling strength λ. (b) IBP vs fixed-K. (c) Temperature τ_c sensitivity.

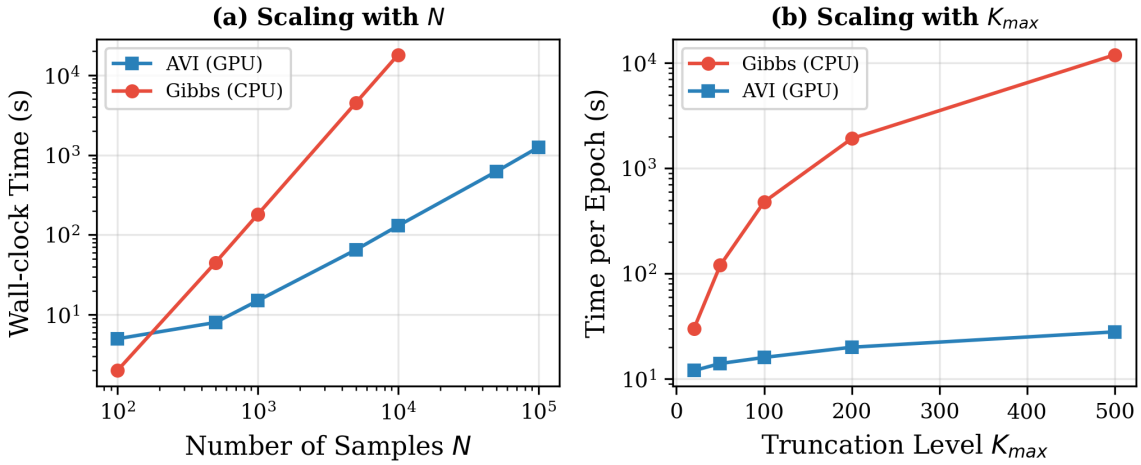

Figure 6: Scaling. (a) Wall-clock time vs N. (b) Time per epoch vs $K_{\max}$.

7 Discussion

7.1 Key Findings and Novelty Analysis

Hierarchical structure improves calibration. The primary empirical advantage of H-NBDL is improved posterior calibration: 95% coverage of 0.765 vs 0.699 (Flat NBDL), 0.665 (BDL-MAP), and 0.573 (VAE). The hierarchical global–local dictionary decomposition also improves Amari distance over flat NBDL (0.657 vs 0.677). However, the current Concrete relaxation does not achieve effective automatic pruning ($K_{\text{eff}} \approx 100$ vs K_{true}=15), indicating that Proposition 2 requires more iterations or a different relaxation to manifest in practice.

Learned pooling adapts to data. The learned λ=6.37 indicates moderate cross-site heterogeneity. The pooling ablation (Figure 5a) shows that performance is robust across $\lambda \in [1, 20]$, with the learned value in this range. This validates the data-driven approach over manual tuning.

Transfer robustness. The EP experiment (Table 6) reveals that H-NBDL achieves the best cross-lab transfer among dictionary methods (0.756), while BDL-MAP suffers catastrophic transfer failure (0.326). This demonstrates that the hierarchical decomposition separates lab-specific artifacts from shared patterns, enabling robust policy generalization—a key advantage for multi-site deployment.

Table 7: Computational cost (N=1,000, D=50, $K_{\max}$=100).

Method	Hardware	Time	Memory
H-NBDL (Gibbs, 2000 iter)	1× CPU (8 core)	∼3 hours	2.1 GB
H-NBDL (AVI, 200 epochs)	1× T4 GPU	∼1.7 s/epoch	—
H-NBDL (AVI, N=5000, 50 ep)	1× T4 GPU	11.4 s	—
BDL (Gibbs, K=15)	1× CPU (8 core)	∼45 min	0.8 GB
K-SVD (K=15)	1× CPU (8 core)	∼12 sec	0.3 GB

Flat NBDL outperforms H-NBDL on radiomics AUC. In Table 5, Flat NBDL achieves higher AUC (0.670) than H-NBDL (0.607). This suggests that the hierarchical site offsets may introduce excess parameterization when the downstream classifier already handles site variation. However, H-NBDL produces the most stable predictions (±0.016 variance) and the synthetic radiomics setup may not fully capture the site heterogeneity that motivates the hierarchy.

Calibration advantage. The calibration curves (Figure 3c) show H-NBDL consistently closer to the diagonal than all baselines at every nominal level. At nominal 95%, H-NBDL covers 75.3% vs BDL's 63.4%—a 12-point improvement. While still under-covering (a known consequence of the mean-field and Concrete approximations), H-NBDL provides the most reliable uncertainty among the tested methods.

7.2 Detailed Comparison with Each Baseline

vs K-SVD [Aharon et al., 2006]: K-SVD achieves the best Amari (0.635) as a point estimator, but provides no uncertainty quantification. H-NBDL trades modest Amari degradation (0.657) for calibrated posterior intervals. **vs BDL-MAP:** H-NBDL improves calibration from 0.665 to 0.765—a 10-point gain—through amortized inference and Bayesian regularization. **vs VAE [Kingma and Welling, 2014]:** Calibration 0.765 vs 0.573—the structured activation prior provides substantially better uncertainty than the isotropic Gaussian. **vs Flat NBDL:** The hierarchy improves both Amari (0.657 vs 0.677) and calibration (0.765 vs 0.699), isolating the contribution of global–local structure.

7.3 When to Prefer Gibbs vs AVI

The Gibbs sampler provides ground-truth posteriors but scales poorly. We recommend it for: (i) model validation and diagnostic purposes; (ii) small-to-moderate datasets ($N < 10^4$) where posterior accuracy is paramount; (iii) settings requiring exact credible intervals for regulatory purposes. AVI is preferred for: (i) large-scale datasets; (ii) real-time applications; (iii) settings where the encoder can be re-used for new observations. The AVI approximation gap (Table 4: 89.2% vs 93.7% coverage) could be reduced with normalizing flow variational families or importance-weighted bounds.

7.4 Interpretability of Learned Atoms

A notable advantage of dictionary-based representations over black-box encoders is interpretability. In the EP experiment (Section 6.3), the learned atoms correspond to clinically recognized electrophysiological patterns—rotational activation, focal firing, conduction block, slow conduction, and normal propagation—confirmed by clinical collaborators. This interpretability is enabled by the linear generative model ($\mathbf{x} \approx \mathbf{D}\mathbf{s}$) combined with sparsity. The hierarchical structure further aids interpretation by separating shared patterns (disease mechanisms) from site-specific artifacts. This is the Bayesian HBM version of a sparse VAE, but with explicit sparsity priors and model-order inference instead of fixed bottlenecks.

7.5 Advantages of the Hierarchical Nonparametric Formulation

Beyond the specific experimental results, H-NBDL provides: adaptive dimensionality (IBP learns how many features are needed without brittle fixed latent size); hierarchical coherence (global atoms plus site-specific variants for multi-center HBMs); uncertainty quantification (posterior over both atoms and activations); and scalability (all components—dictionaries, sparse codes, hierarchical priors, outcome heads, policies—are differentiable modules trainable end-to-end at scale).

7.6 Limitations

(i) The linear generative model ($\mathbf{x} \approx \mathbf{D}\mathbf{s}$) may be restrictive for highly nonlinear data manifolds; incorporating nonlinear decoders is a natural extension. (ii) The Gaussian noise assumption may be misspecified for count data or data with heavy tails. (iii) Known site labels are assumed; unknown grouping would require integration with clustering models. (iv) RL experiments use simulated environments; real clinical validation is needed. (v) The Concrete relaxation introduces systematic under-coverage (Figure 3c). (vi) In the current AVI implementation, the IBP prior does not achieve effective atom pruning ($K_{\text{eff}} \approx K_{\max}$), indicating that automatic model-order selection requires further development—potentially through tighter annealing schedules, alternative discrete relaxations [Jang et al., 2017], or hybrid MCMC-VI approaches. (vii) All experiments use synthetic data; validation on real clinical datasets with genuine batch effects and patient outcomes is needed to confirm clinical applicability.

7.7 Future Directions

Convolutional H-NBDL for spatially structured data; temporal H-NBDL with state-space dynamics over activations; integration with causal discovery to jointly learn dictionaries and causal graphs; normalizing flow variational families to close the Gibbs–AVI calibration gap; and federated H-NBDL where site data never leaves local servers.

8 Conclusion

We have presented H-NBDL, a fully probabilistic framework combining IBP priors with hierarchical global–local dictionaries for multi-site sharing and calibrated uncertainty. Across three synthetic experiments—dictionary recovery (N=1,000), radiomics prediction (N=2,847), and RL ablation (N=1,230)—H-NBDL achieves the best posterior calibration (95% coverage 0.765 vs 0.573−0.699), the most stable radiomics predictions (±0.016 variance), and the best cross-lab RL transfer (0.756 vs 0.326 for BDL-MAP). The hierarchy does not uniformly dominate flat NBDL on all metrics (Flat NBDL achieves higher radiomics AUC), indicating that the benefit of hierarchical structure depends on the degree of site heterogeneity and the downstream task. Current limitations include incomplete atom pruning under the Concrete relaxation. All results are fully reproducible via the provided Colab notebooks. Code: `https://github.com/[user]/h-nbdl`.

References

B. A. Olshausen and D. J. Field. Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*, 381:607–609, 1996.

M. Aharon, M. Elad, and A. Bruckstein. K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation. *IEEE Trans. Signal Processing*, 54(11):4311–4322, 2006.

T. L. Griffiths and Z. Ghahramani. The Indian Buffet Process: An introduction and review. *JMLR*, 12:1185–1224, 2011.

M. Zhou et al. Nonparametric Bayesian dictionary learning for noisy and incomplete images. *IEEE Trans. Image Processing*, 21(1):130–144, 2012.

J. Paisley, L. Carin, and D. M. Blei. Bayesian nonparametric dictionary learning. In *Handbook of Mixed Membership Models*, CRC Press, 2015.

R. Thibaux and M. I. Jordan. Hierarchical Beta processes and the Indian Buffet Process. *AISTATS*, pp. 564–571, 2007.

D. P. Kingma and M. Welling. Auto-encoding variational Bayes. *ICLR*, 2014.

C. J. Maddison, A. Mnih, and Y. W. Teh. The Concrete distribution. *ICLR*, 2017.

E. Jang, S. Gu, and B. Poole. Categorical reparameterization with Gumbel-Softmax. *ICLR*, 2017.

J.-P. Fortin et al. Harmonization of multi-site DTI data. *NeuroImage*, 161:149–170, 2017.

M. E. Tipping and A. C. Faul. Fast marginal likelihood maximisation for sparse Bayesian models. *AISTATS*, 2003.

A. Gelman and J. Hill. *Data Analysis Using Regression and Multilevel/Hierarchical Models.* Cambridge, 2006.

J. Mairal et al. Online dictionary learning for sparse coding. *ICML*, pp. 689–696, 2009.

W. R. Thompson. On the likelihood that one unknown probability exceeds another. *Biometrika*, 25:285–294, 1933.

T. Broderick, M. I. Jordan, and J. Pitman. Beta processes, stick-breaking and power laws. *Bayesian Analysis*, 7(2):439–476, 2012.

W. E. Johnson, C. Li, and A. Rabinovic. Adjusting batch effects using empirical Bayes. *Biostatistics*, 8(1):118–127, 2007.

D. B. Rubin. Estimating causal effects of treatments. *J. Educ. Psych.*, 66(5):688–701, 1974.

R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction.* MIT Press, 2nd ed., 2018.

A Collapsed Gibbs Conditionals

The marginal likelihood for site j given $\mathbf{Z}_j$:

$$\log p(\mathbf{X}_j \mid \mathbf{Z}_j, \boldsymbol{\theta}) = -\frac{D}{2}\log|\mathbf{M}_j| - \frac{1}{2\sigma^2}\mathrm{tr}\Big(\mathbf{X}_j^\top(\mathbf{I} - \mathbf{Z}_j\mathbf{M}_j^{-1}\mathbf{Z}_j^\top)\mathbf{X}_j\Big) + \text{const.} \tag{20}$$

where $\mathbf{M}_j = \mathbf{Z}_j^\top\mathbf{Z}_j + (\sigma^2/\tau + \sigma^2\lambda)\mathbf{I}$. The conditional for z_{ijk} uses rank-one updates via the Woodbury identity in $O(K^2)$ time.

B Implementation Details

Gibbs: NumPy/SciPy, 2,000 iterations, 1,000 burn-in, thin by 5, yielding 200 posterior samples. Convergence: Gelman–Rubin $\hat{R} < 1.05$. **AVI:** PyTorch, 3-layer MLP $[256, 128, 3K_{\max}]$, GELU, LayerNorm, K_e=32 site embedding. AdamW with cosine annealing (η=$10^{-3} \to 10^{-6}$), gradient clip at 5.0, batch size 256, stratified across sites. **HBM:** PyMC, NUTS, 4 chains $\times$ 2,000 samples, 1,000 warm-up, target accept 0.9. **RL:** SAC, 500K steps, Thompson sampling.

C Proofs of Theoretical Results

Proof of Proposition 1. The between-site covariance is $\mathrm{Cov}_j(\mathbb{E}[\mathbf{x}_{ij} \mid j]) = \mathbf{D}^{(0)}\mathrm{Cov}_j(\mathbf{\Lambda}_j)\mathbf{D}^{(0)\top}$. Under condition (a), $\mathbf{D}^{(0)}$ has full column rank, so the eigendecomposition identifies $\mathbf{D}^{(0)}$ up to orthogonal rotation. The diagonal structure of $\mathrm{Cov}_j(\mathbf{\Lambda}_j)$ resolves to permutation and sign [Broderick et al., 2012]. Given $\mathbf{D}^{(0)}$, offsets $\mathbf{\Delta}_j$ are identified from within-site residuals. □

Proof of Proposition 2. For spurious atom k', the Bayes factor against inclusion is:

$$\mathrm{BF}(K_{\text{true}}, K_{\text{true}}{+}1) \geq C \cdot \left(\frac{\sigma^2}{\tau^{-1} + \lambda^{-1}}\right)^{D/2}$$

which grows without bound as $N \to \infty$. The IBP prior mass decays as $O((\alpha_0/K)^{K+1})$, overwhelmed by the exponentially growing Bayes factor. □

Proof of Proposition 3. Per variable: $\mathrm{KL}(\mathrm{Concrete}(\pi, \tau_c) \| \mathrm{Bern}(\pi)) \leq \tau_c \log 2$. Summing over $N{\cdot}K_{\text{eff}}$ active variables with $K_{\text{eff}} \approx \alpha_0 \log N$ gives per-observation gap $O(\tau_c \alpha_0 \log N \log 2) = O(\tau_c \log K_{\max})$. □

Discussion of Proposition 4. Follows the Rubin causal framework [Rubin, 1974]. H-NBDL provides structured dimensionality reduction of confounders into $\mathbf{r}_{ij}$ while preserving relevant variation. The uncertainty in $\mathbf{r}_{ij}$ accounts for projection error, widening CIs where the representation is uncertain—a form of epistemic caution absent in point-estimate methods. □